YISHUN INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

Suggested answer

Section A

Structured Questions

Answer both questions in this Section

Cluster 4 : Fieldwork

1 A group of ten 18-year-old students from Crystalis Junior College will visit Freiburg from 1–10 December to investigate whether Freiburg's sustainability strategies are transferable to Singapore's high-density urban context.

Freiburg is widely recognised as Germany's "green city" and is regarded as a leading example of urban sustainability and renewable energy transition. In their background research, the students found that approximately 60% of Germany's electricity is generated from renewable sources, compared to about 5% in Singapore. They also learnt that the Vauban district is well known for its sustainable transport system, low car ownership and widespread use of rooftop solar photovoltaic (PV) systems.

The students developed the following hypothesis:

High levels of resident support for renewable energy and sustainable transport strategies contribute significantly to the success of Freiburg's sustainable urban model.

To investigate this hypothesis, the students will conduct a questionnaire survey of 40 residents in the Vauban district of Freiburg using convenience sampling. The survey will examine residents' support for renewable energy, including their willingness to adopt rooftop solar panels, as well as their use of sustainable transport. The students will also carry out an on-site field inspection to observe renewable energy installations and sustainable transport infrastructure.

Resource 1 shows a map of the fieldwork site in the Vauban district. Resource 2 shows photographs taken during the on-site field inspection. Resource 3 provides the questionnaire used by the students. Resource 4 shows the residents' responses to selected survey questions.

(a) Explain the strengths and weaknesses of the students' hypothesis. [4]

Award 1m for each relevant point from any of the below 2 dimensions. Need to consider both the strengths and weaknesses.

- [**At suitable scale**] The students have devoted sufficient number of days (10 days) to the investigations and also enough manpower (10 students) to complete the primary investigation in the allocated time. (1)

- [**researchable or measurable**] The results for the research question can be addressed using primary study of 40 residents in the Vauban district of Freiburg. The survey can elicit the response for the residents' support for renewable energy, including their willingness to adopt rooftop solar panels, as well as their use of sustainable transport. In addition, the students will also carry out an on-site field inspection to observe renewable energy installations and sustainable transport infrastructure. (1-2)

- [**Clearly defined**] e.g, The target group of "*resident"* is clearly defined. (1) But could have been more clear as to what constitute 'high level of resident support' (1)

(b) Explain the advantages and limitations of using convenience sampling for this investigation. [4]

Award 1m for each relevant point. Must consider both advantages and limitations.

Convenience sampling is
- q**uick and practical** – researchers can approach respondents in pedestrian areas, seating areas, cafés and public spaces. (1)

It is easier to obtain responses from **participants who are comfortable answering the questionnaire**. It is appropriate for an exploratory investigation where the aim is to identify **general trends and perceptions**. (1)
But it does not stratified the respondents such that it is representative of the demographic in the areas. (1) Other issues include s**election bias** – people who are present at the site and willing to respond may not represent all elderly people. And l**ocation bias** – surveying only people in busy or accessible parts of the area may exclude some groups of residents. (1)

(c) Explain how an on-site field inspection of the Vauban district provides evidence that complements the survey findings when investigating the students' hypothesis. [5]

Award 1m for each relevant point. Award 1 additional mark for further development of each explanation.
As this is the **first time** at the site, conducting a **site reconnaissance or on site field inspection** is useful for students to obtain **primary data** on the eco-friendly features in the estate. [1 mark] For example, students are able to observe that there are use of solar panels and trams [1 additional mark]

Based on their observations, this could help students determine what other relevant primary data to collect, such as to count the number of homes with solar panels. [1 mark]

The students can also use the information to better plan their investigation, **ask relevant questions** about sustainable living habits in their survey. [1 mark]. For example, based on the features observed, the students can ask the residents about their opinions on the preference for similar features [1 additional mark].

(d) With reference to Resource 3, suggest and justify improvements to the survey questionnaire. [5]

Award 1m for each relevant point. Award 1 additional mark for further development of each explanation.

- **Reduce the number of sensitive questions on residents' profiles**, such as household income and the number of household members. (1)These questions may be intrusive and make respondents uncomfortable, resulting in **non-response or inaccurate answers**. So need to reduce the sensitive questions and include only relevant demographic questions should be retained. (+1) **Or provide a "Prefer not to say" option for sensitive demographic questions**, such as income or age. This gives respondents greater privacy and may encourage more people to **complete the questionnaire honestly**. (+1)
-
- **Reorder the Likert scale so that "Strongly Agree" is scored 5 and "Strongly Disagree" is scored 1.** (1) This is more intuitive and reduces confusion when respondents interpret and complete the questionnaire, improving the **reliability of the responses**. (+1)
- **Include open-ended questions** to allow residents to elaborate on their experiences and opinions. (1) This can provide **qualitative information and**

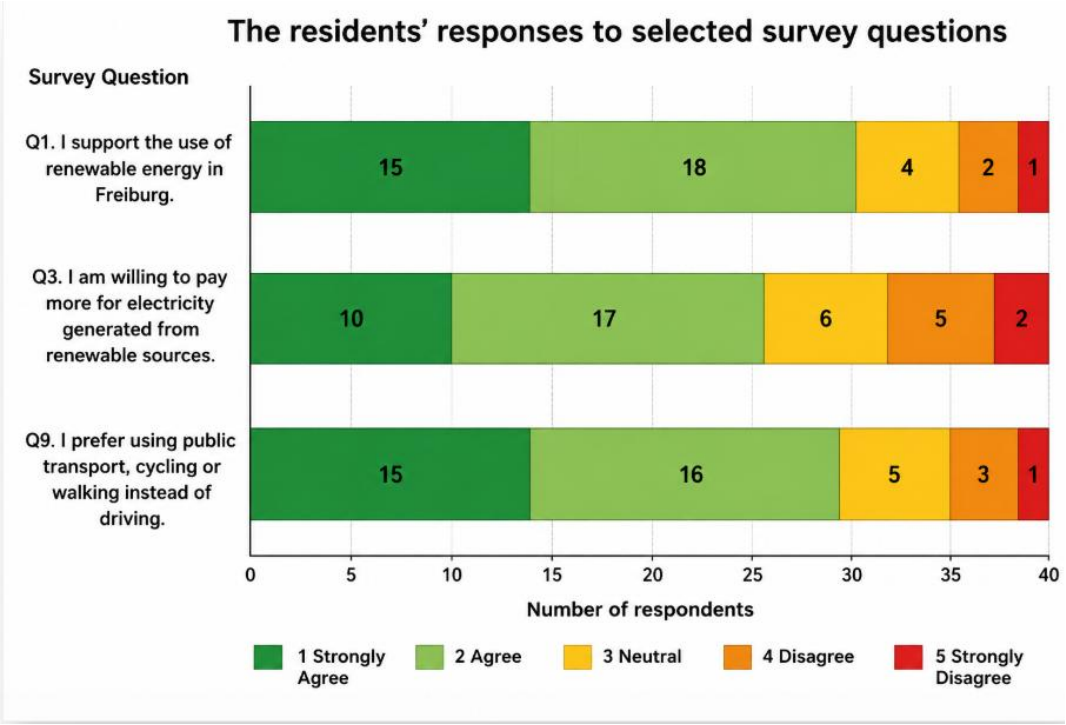
detailed explanations that cannot be captured through fixed-response questions. (+1)

- **Avoid redundant question such as Question 7.** Public transport is convenient for my daily travel. **(1)** This would make the questionnaire **shorter and less tiring for respondents**, reducing respondent fatigue and improving the quality and completeness of the data collected. (+1)

(e) Explain how the students could process and present the survey data in Resource 4 to facilitate analysis of their hypothesis. [5]

Stacked bar chart is preferred because:

- **It allowed the responses from the three questions to be compared easily**.
- Responses can be grouped into **positive, neutral and negative attitudes**.
- The relative proportion of respondents with different attitudes can be seen clearly.
- Contrasting colours or grduational colours can be used to show the difference. And the 'positive' response is likely to stand out in all 3 questions. For instance, Q1 has **33 out of 40 respondents (82.5%)** who either strongly agree or agree that they support renewable energy. A diverging stacked bar chart would make this strong positive response immediately visible.



(f) Evaluate the extent to which the data collection methods used in this investigation enabled the students to test their hypothesis effectively. [10]

Judgement

Overall, the data collection methods enabled the students to **test the hypothesis to a moderate extent**. The questionnaire and on-site field inspection provided evidence of **residents' perceptions and observable sustainability features**. However, **convenience sampling, a small sample size and possible language barriers** reduced validity. More importantly, the methods could identify an association between resident support and Freiburg's sustainability, but **could not establish that resident support led to its success**.

Advantages

The **questionnaire survey** was appropriate because the hypothesis concerns **resident support** for renewable energy and sustainable transport. Likert-scale questions allowed attitudes to be quantified, while questions on willingness to install solar panels or use public transport linked attitudes to intended behaviour. Using the same questionnaire also improved **consistency and comparability**.

The **on-site field inspection** provided useful **triangulation**. Students could observe solar panels, tram infrastructure, pedestrian facilities and low-car environments, allowing questionnaire responses to be compared with actual sustainability features rather than relying solely on residents' perceptions.

Limitations

However, **convenience sampling** reduced representativeness because students surveyed residents who were easiest to access, such as those at tram stops and public spaces. Such residents may be more supportive of sustainable transport, creating **selection bias**.

Conducting fieldwork in a **foreign country** also created challenges. Students' unf

amiliarity with Vauban may have encouraged sampling at easily accessible locations. **Language barriers** could further bias the sample towards residents able to communicate in English. The relatively **small sample size** also limits how confidently findings can be generalised to Vauban's population. Responses may additionally be affected by **social desirability bias**, especially for environmentally responsible behaviour.

Most importantly, the questionnaire measures **resident support**, whereas the hypothesis claims this contributes significantly to Freiburg's success. It cannot

establish **causation**. Resident support may instead result from successful policies, while government planning, investment, infrastructure and financial incentives may be equally important.

Improvements

Students could use **stratified or systematic sampling** across different parts of Vauban and survey residents at different times. Providing questionnaires in **German and English** would reduce language bias, while a larger sample would improve representativeness. Secondary data on **solar-panel adoption, car ownership and public transport use**, together with interviews with residents and planners, would better link stated support to actual sustainability outcomes.

There is room to consider other primary research including conducting interviews with the authorities (such as solar panels company or the local government) to ascertain the take-up rate for the sustainability measures.

Overall Judgement

Therefore, the methods were effective in identifying **levels of resident support**, but sampling weaknesses and the inability to establish causation limit the conclusion. They provide **some support for the hypothesis**, but stronger and more representative evidence is needed to show that resident support significantly contributes to Freiburg's sustainable urban model.

(g) Explain how you would design an extension to this investigation to determine whether Freiburg's sustainability strategies are transferable to Singapore. [7]

The investigation could be extended to determine whether Freiburg's sustainability strategies are transferable to Singapore through pre-fieldwork, fieldwork and post-fieldwork stages.

Begin with a Hypothesis/Research Question:
The investigation could ask: To what extent can Freiburg's sustainability strategies be transferred and adapted to Singapore?" The hypothesis could be that Freiburg's sustainability strategies can be transferred to Singapore, but their effectiveness will vary due to differences in physical, social and economic conditions.

Other research could include Freiburg's key strategies, such as cycling infrastructure, public transport, renewable energy and sustainable housing, and identify comparable strategies in Singapore. Secondary data such as cycling modal share, public transport usage, energy consumption and carbon emissions could be collected for both cities. Sampling locations in Singapore should be selected to represent areas with different levels of sustainable infrastructure.

During the fieldwork phase, conduct survey questionnaires with residents in Singapore to investigate attitudes towards and usage of sustainable transport and energy strategies. Environmental quality surveys, and pedestrian/cyclist counts could also be conducted at selected sites. Stratified sampling could ensure that different residential areas and socio-economic groups are represented.

During the post-fieldwork, present the data using comparative bar graphs, maps and scatter graphs. Compare the effectiveness and public acceptance of Freiburg's strategies with those in Singapore. Statistical techniques such as Spearman's rank could be used to test co-relationships between accessibility and usage.

Evaluation

The results should be evaluated by considering differences in **climate, population density, land availability, government policies, infrastructure and cultural attitudes**. This would allow a conclusion to be made about which Freiburg strategies could be directly transferred and which would need to be **modified to suit Singapore's context**. Limitations such as questionnaire bias, sampling size and differences between the two cities should also be considered.

Levels	Marks	Generic Level Descriptors for H2 Fieldwork Evaluative Question
3	9–10	Evaluation is analytical and coherent. Response addresses the question and demonstrates good knowledge and understanding of fieldwork methods relevant to the given context.
2	5–8	Evaluation is mostly analytical and coherent. Weaker responses in this level will have evaluation that is broadly analytical and generally coherent. Response generally addresses the question and demonstrates adequate knowledge and understanding of fieldwork methods relevant to the given context.
1	1–4	Response is descriptive with limited or no analysis and evaluation. Response is fragmented and lacks clarity. Response lacks focus on the question and may be largely irrelevant to it. Response shows basic knowledge and understanding of fieldwork methods. Response has some, though limited, relevance to the given context.
0	0	No creditworthy response

Cluster 3 : Sustainable Future and Climate Change

2 Resource 5 shows the stages of reimaging in selected Melbourne suburbs. Resource 6 shows the indicators of reimaging in selected suburbs in Melbourne. Resource 7 Footscray before and after urban renewal. Resource 8 is an Australian newspaper report on gentrification in Melbourne.

(a) Using Resource 5, describe the spatial pattern of urban reimaging in Melbourne. [5]

Suggested answer

The spatial pattern is one of a "ripple effect", with the most established urban reimaging concentrated around the inner city and progressively newer stages occurring further from the CBD. (1) The pattern is not perfectly concentric. There is a noticeable westward extension towards Footscray, Yarraville and Braybrook, and a northern corridor towards Brunswick, Thornbury and Preston. (This suggests that urban reimaging is influenced by transport routes and

accessibility, as many of these suburbs are located along major roads and railway lines.) (+1)

The **strongest and most advanced reimaging** occurs close to the CBD, particularly in the **inner-northern and inner-eastern** suburbs of Fitzroy, Collingwood and Richmond, which are classified as mature/advanced gentrification dating from the 1990s. (1)

Moving further from the CBD, the intensity generally declines and the timing becomes more recent. Brunswick and Footscray, located further north and west, show advanced/ongoing gentrification from the 2000s–2010s. (1)

Further out, **suburbs** such as Preston, Yarraville and Braybrook show emerging or **early-stage gentrification**, mainly from the 2010s–2020s. This indicates that urban reimaging is spreading outward from the established inner-city areas. Beyond these areas, many suburbs such as Reservoir, Ivanhoe, Hawthorn, Malvern and Caulfield are classified as having early/potential gentrification, suggesting that they have experienced less change but may be affected in the future. (+1)

Examiners' comment
Weakest answer can only count the number of areas with early Vs more advanced stage of gentrification. These are also the ones who failed to note that there is a pattern in the advancement of gentrification over time. Others lost marks for not identifying the names of cities near the CBD which have experienced mature gentrication and those further away from the cities which have only early gentrification.
Very few can see that there is a progressive advancement in the Northern/ Northeastern and western directions. In addition, the inability to identify anomaly also costs marks. The main idea of General - Specific – Anomaly is not practised. (Aka GARA – Generic, Absolute, Relative, Anomaly)

(b) Using Resource 6, explain how the demographic and housing data provide evidence that urban reimaging has occurred in the suburbs. [4]

Suggested answer
Correctly identified 2 points and their correlations to gentrification. Plus provide examples to illustrate.

The demographic data provides evidence of **gentrification and changing socio-economic characteristics**.

Demographic - Those with bachelor's degree or higher and those employed in professional occupations has increased. (1) For

example, in Fitzroy, residents with a bachelor's degree or higher increased from **28% in 2000 to 58% in 2024**, while those employed in professional occupations increased from **24% to 46%**.Similar increases occurred in Footscray and Braybrook, suggesting an influx of more highly educated and professional residents. (+1)

Housing - The housing data also indicates **increased demand and investment following urban reimaging**. (1) Median house prices in Fitzroy rose from **A$0.45 million to A$1.85 million**, while median weekly rent increased from **A$230 to A$760** between 2000 and 2024. (+1)

Examiners' comments
The question mentioned housing and demographic evidence. Hence ideally you should have picked data related to housing and demographic properties. The vacant/industrial land may not fall into such categories.
Nonetheless, marks are subsequently also given for identification of such evidence as vacant/industrial land which fell from 18% to 6%, for it can suggest the conversion of underused land into residential and commercial uses.

(c) Using Resource 7, describe the physical changes that have taken place following urban reimaging in Footscray. [4]

Suggested answer

- **Land use changed** from large, vacant/underused industrial buildings in 2000 to **mixed-use developments**, including shops, cafés and restaurants in 2024.
- **Buildings were redeveloped** from old, weathered brick industrial warehouses with graffiti to **modern, well-maintained brick residential and commercial buildings**.
- **Greenery increased significantly**, with the previously largely bare and concrete landscape replaced by **mature trees, shrubs, planted areas and landscaped spaces**.
- **The public realm became more pedestrian-friendly**, with the former wide, empty industrial access area transformed into **paved pedestrian streets, seating areas, outdoor dining spaces and better-designed public spaces**.

Examiners' comments
Though this is a 'describe' question, using comparative skill will have brought out the idea of changes over time. Otherwise you could have shown how there has been an increased in greenery, increase maintenance, increased F&B etc to illustrate the change.

(d) Using Resource 8, explain how urban reimaging can affect community identity in Footscray. [5]

Suggested answer
The impact should be both positive and negative.
(+) Reimaging can change **the social composition of the community** as new, younger and more affluent professionals move into Footscray. The growth of young professionals mentioned by **café owner Daniel Nguyen can introduce new lifestyles and values**, changing the traditional character of the neighbourhood. (1)
(+/-) **Changes in businesses and land use can alter the local identity of the area**. (1) Former industrial sites and warehouses have been replaced by apartments, cafés, restaurants and mixed-use developments, creating a more vibrant and commercially oriented identity. (+1)

(-) Urban reimaging can weaken existing community identity through **displacement**. Rising rents and property prices have made it difficult for lower-income households and long-term residents to remain. **Maria Lopez** noted that many families who had helped build the neighbourhood could no longer afford to stay, potentially weakening established social networks and community ties.(1)

(+) However, reimaging can also strengthen community identity by creating **improved public spaces, community parks, cycling paths and streetscapes** where residents can interact. These improvements can **encourage greater social interaction and create a stronger sense of place** among both existing and new residents.(1)

(e) Explain how urban reimaging can create both complementarity and tensions between sustainable urban development and liveability. [6]

Suggested answer
Award 1m for each relevant point. Award 1 additional mark for further development of each explanation.

- **Green infrastructure → complementarity**: Redeveloping brownfield sites such as derelict land into parks, green spaces and cycling paths can improve environmental sustainability while also increasing residents' access to recreation and improving liveability. (1)
- **Improved public transport → complementarity**: Investment in public transport and pedestrian infrastructure can reduce car dependence and carbon emissions while making the city more accessible and convenient for residents. (1)

- **Higher-density mixed-use** development → **tension**: Building apartments near shops and services can reduce urban sprawl and encourage walking, but **increased density may cause congestion**, noise and overcrowding, reducing liveability for some residents. (1)
- **Urban renewal → tension**: Reimaging can improve the appearance and economic vitality of declining areas, but **rising inflationary pressure and property prices/rents** may cause gentrification and displacement, making the area less affordable for existing residents. (1)
- **Economic development → complementarity and tension**: New cafés, businesses and mixed-use developments can create **jobs and services**, improving liveability and economic sustainability. However, **commercialisation** may change the character of neighbourhoods and make them less accessible to lower-income residents. (1)

  Overall: Therefore, urban reimaging can create strong complementarity where environmental improvements also benefit residents, but tensions arise when sustainability and economic objectives increase costs or reduce accessibility for certain social groups.

(f) Explain how government can make Footscray more liveable for the women. [6]

Suggested answer

Award 1m for each relevant point. Award 1 additional mark for further development of each point with explanation/examples.

Since Footscray is undergoing urban reimaging, consider how it attracts a more diverse population.

1. **Improve public transport and safety** – Government can provide frequent and reliable trains, buses and trams, together with better lighting, CCTV and safe pedestrian routes. This can improve women's perception of safety, particularly for those travelling alone at night or commuting after work.
2. **Provide affordable childcare** – As Footscray attracts more working families, the government can support affordable childcare centres and childcare facilities near transport hubs. This can reduce the time and financial pressures faced by ***working mothers*** and help them balance employment with family responsibilities.
3. **Develop accessible healthcare services** – More clinics, women's health centres and accessible healthcare facilities can be provided. This would meet the different needs of women across life stages, including *younger women, mothers and elderly women*, improving their overall quality of life.

4. **Create safe and inclusive public spaces** – Parks, plazas and recreational areas should have good lighting, seating, toilets and accessible pathways. These spaces can support social interaction and physical activity, benefiting both *younger women and older women* who may otherwise experience social isolation.
5. **Provide affordable housing** – Government can increase the supply of affordable and social housing as urban reimaging may increase property prices and rents. This is important for *single women, lower-income women and elderly women*, reducing the risk of displacement caused by gentrification.
6. **Reduce work–life pressures** through mixed-use planning – Government can encourage housing, workplaces, shops, schools, childcare and services to be located close together. This reduces **commuting and time pressures**, particularly for working women who may have to manage both paid employment and household or caregiving responsibilities.

Overall judgement: Government should adopt a gender-sensitive approach to urban reimaging, rather than assuming all women have the same needs. Younger working women may prioritise transport, safety and childcare, while older women may place greater importance on healthcare, accessibility, affordable housing and social spaces.

Examiners' comments
Marks are lost for narrow focus. For instance, some could only discuss infrastructural improvement for women, from well-lit streets to wider walkways for prams, or generic improvements in cleanliness and pay which benefits all walks of life

Section B

Answer **EITHER** question 3 **OR** question 4.

Examiners' comments

1. Poor essay structure

Many responses have weak or incomplete essay structure, ranging from limited paragraphing to the absence of a clear introduction and conclusion.

A good introduction should establish your overall position or line of argument. If you begin writing without making your stand clear, the examiner is left uncertain about where your argument is heading.

Likewise, a conclusion is not optional. It should return to the question, weigh the arguments presented and provide a clear, justified overall judgement. An essay without a conclusion often gives the impression that the argument has not been fully resolved.

Think before you write: What is my overall stand? What are the main arguments that will support it? What factors might challenge my stand?

2. Weak or unclear topic sentences

The first sentence of each paragraph should communicate the main argument of that paragraph. Think of the topic sentence as the mini-answer to the question.

A weak or overly general opening forces the examiner to read through the entire paragraph to figure out what you are trying to argue. This makes your line of reasoning difficult to follow.

A strong topic sentence should therefore:

-directly address the question;
-identify the factor or argument being discussed; andwhere appropriate, indicate the direction of the argument.

Do not make the examiner go on a treasure hunt to find your argument. Tell them what you are arguing first, then prove it.

3. Contradictory conclusion

Some essays end with a conclusion that contradicts the stand established in the introduction.

This suggests that the overall argument has not been thought through carefully. Before you begin writing, take a moment to decide what you actually believe the evidence shows.

Your conclusion should not introduce a completely different position. It should be the logical outcome of the arguments and evaluation developed throughout the essay.

A useful check is:

Does my conclusion answer the exact question, reflect my arguments, and match my overall stand?

4. Misunderstanding what a 20-mark essay requires

The most significant issue is that many students do not fully understand what is required of a 20-mark essay.

A 20-mark essay is not simply an exercise in explaining geographical factors. Neither is it a matter of merely agreeing or disagreeing with the statement.

A strong 20-mark essay needs to demonstrate:

Knowledge → Explanation → Application → Evaluation → Judgement

You need to consider both sides of the argument, explain why each side may be valid, and then evaluate the relative importance of the arguments before arriving at a justified judgement.

In other words:

Do not just tell me what happens. Tell me why it happens, WHY it matters, HOW important it is, and under what **circumstances** your argument holds true.

Levels	Marks	Generic Level Descriptors for H2 Essays
5	18–20	Evaluation is consistently analytical and coherent. Response is well-supported by relevant material, including the effective use of examples. Response features accurate geographical knowledge and reflects good understanding of the subject content relevant to the question.
4	14–17	Evaluation is analytical and coherent. Response is mostly well-supported by relevant material, including the appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
3	10–13	Evaluation is broadly analytical and generally coherent. Response is moderately well-supported by relevant material, including some appropriate use of examples. Response features accurate geographical knowledge and reflects adequate understanding of the subject content relevant to the question.
2	6–9	Response is largely descriptive with limited analysis and evaluation. Response is partly coherent and may lack clarity in parts. Response is poorly supported by relevant materials, including the limited use of examples. Response features inaccurate geographical knowledge and poor understanding of the subject content relevant to the question.
1	1–5	Response is descriptive with no analysis or evaluation. Response is fragmented and lacks clarity. Response consists of unsupported assertions. Response features largely inaccurate geographical knowledge and a lack of understanding of the subject content relevant to the question.
0	0	No creditworthy response

Cluster 3 : Sustainable Future and Climate Change

EITHER

3 Economic development improves urban liveability, but also creates waste management challenges." Evaluate the validity of this statement. [20]

The essay should argue that the statement is largely valid, but the relationship between economic development, liveability and waste is not inevitable. The key is to evaluate how the benefits of development can be balanced against its environmental costs. Establish the overall argument that economic development initially improves liveability but can generate increasing waste through consumerism and commercialisation. However, wealthier cities also have greater capacity to manage these challenges.

Paragraph 1 – Economic development **improves liveability as well as creates waste management challenges for some**. Explain how economic development generates employment, higher incomes and government revenue, allowing investment in housing, healthcare, transport, sanitation and public infrastructure. These improve economic, social and environmental dimensions of liveability. Evaluate by noting that such benefits creates waste challenges for some. Rising incomes increase consumerism, while commercialisation through

malls, restaurants, e-commerce and businesses generates greater quantities of packaging, food and electronic waste. **High-income countries** can therefore produce significantly more waste per person. This creates challenges of waste disposal, including landfill capacity, pollution and incineration. Examples of Hong Kong dumping its waste in China and the multi-year diplomatic conflict between Canada and The Philippines triggered by a private commercial shipment of garbage to The Philippines.

Paragraph 2 : Economic growth can come at the expense of urban liveability as well as aggravates the waste problems. The examples of crowded cities with waste issues in LICs, e.g., Dharavi and brown issues.

Paragraph 3 –: Evaluate whether these **waste challenges can be overcome from waste disposal to reduction and prevention**. Wealthier cities have greater financial and technological capacity for recycling, waste-to-energy and efficient collection. However, these mainly manage waste after it is produced. Greater emphasis should therefore be placed on waste reduction and prevention, including circular economy approaches, reuse and product redesign. Consider the various strategies to deal with the waste issues in HICs which the LICs may not have the luxury to do so.

Paragraph 4 – Consider how more can be done by firms and governance to allow for **improved urban liveability without the negative effects of waste management challenges**. While businesses can develop greener production, greenwashing may undermine sustainability. Strong government regulation, extended producer responsibility and controls on packaging are therefore necessary. This is unique to the high income countries so relevant examples can be drawn from the high income countries.

Stand : Economic development improves liveability most sustainably when economic growth is decoupled from resource consumption and waste generation. Thus, the outcome depends less on development itself and more on governance, consumption patterns and the adoption of circular-economy strategies.

OR

4 Evaluate the extent to which adaptation strategies should be prioritised over mitigation strategies in responding to climate change. [20]

The essay should argue that adaptation should be prioritised to a significant extent because it addresses the **immediate and pressing impacts** of climate change, but mitigation remains essential for reducing the scale of **future climate change**. This is because mitigation measures involve reducing greenhouse gas emissions or enhancing carbon sinks, and adaptation as adjusting to actual or expected climate impacts. Establish that adaptation provides more immediate benefits, while mitigation addresses the root cause of climate change. The extent to which either should be prioritised depends on a country's development and vulnerability. In addition, the HICs can take the lead by prioritising the mitigation strategies since they have the financial capacity to

wait out the outcome of the mitigation strategies since they will most likely have the advantage of technological advancement to put in place their adaptation strategies.

Paragraph 1 – Importance of adaptation: Explain how adaptation can directly reduce current risks from extreme climate events such as flooding, droughts, heatwaves, sea-level rise and food insecurity (Note that climate change is not equivalent to flooding). Measures such as flood barriers, early-warning systems, drought-resistant crops and heat shelters can protect lives and livelihoods. This is particularly important for low-income countries and vulnerable communities with limited capacity to cope with climate hazards. E.g., Small island developing countries are left with little choices but to adapt to climate change.

Yet, mitigation remains essential: Mitigation tackles the underlying cause by reducing emissions through renewable energy, energy efficiency, public transport and carbon capture. Without mitigation, climate change will continue to intensify, increasing the scale and cost of adaptation. Therefore, adaptation alone cannot provide a long-term solution. Given the gravity of the climate change, it is not enough to react to climate change. Instead, there is a need to seek out solutions to nip it in the bud and mitigate the source of the problem, ie., the green house gases. (Do not at this point went into great details on the human Vs natural causes of climate change).

Paragraph 2 – Differing priorities by income : High-income countries should arguably place greater emphasis on mitigation because they have greater financial, technological and institutional capacity and historically contributed more to emissions. In contrast, low-income countries may reasonably prioritise adaptation because they are often more vulnerable and have contributed less to cumulative emissions. (do note that adaptation strategies despite the financial costs are not as expensive compared to the mitigation strategies which prove time-consuming and may not see results). Rich countries take the lead in partnership on climate action. Norway is the world's largest financial backer of REDD+ (Reducing Emissions from Deforestation and Forest Degradation) through its International Climate and Forest Initiative (NICFI), having committed billions of dollars to curb global tropical deforestation. In Indonesia, Norway has pledged a $1 billion REDD deal to help fight deforestation. (https://news.mongabay.com/2022/09/indonesia-and-norway-give-redd-deal-another-go-after-earlier-breakup/)



Paragraph 3 – Need for a combined approach: Evaluate the false dichotomy between the two. High-income countries can support adaptation finance and technology transfer to vulnerable countries while reducing their own emissions. Examples of how Singapore participates in both adaptation, without neglecting mitigation strategies.

Stand: Adaptation should be prioritised in the short term, particularly in highly vulnerable countries, because it addresses immediate impacts. However, mitigation should receive greater priority in the long term, especially from high-income countries, as continued emissions will make adaptation increasingly difficult and expensive. Therefore, the most effective response is a differentiated but complementary approach rather than prioritising one universally.

[Total : 90 marks]

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